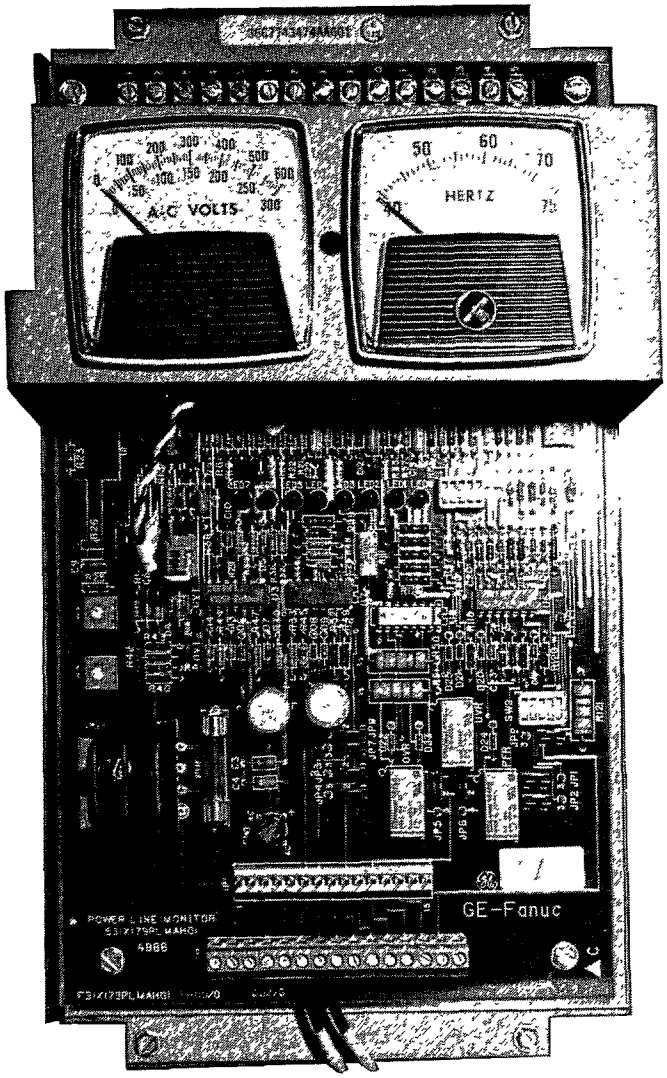
GE DRIVE SYSTEMS

INSTRUCTIONS



POWER LINE MONITOR

CAUTION: To ensure proper operation of the equipment, these instructions must be followed for proper set-up and installation of this device.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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SAFETY PRECAUTIONS

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.
WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to or destruction of the equipment.
CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.
NOTE labels will be white in color with black lettering.

WARNINGS, CAUTIONS AND NOTES IN THESE INSTRUCTIONS

The following paragraphs list some general safety reminders and safety recommendations to be followed when installing or operating this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - PERSONAL INJURY HAZARD: Always read the complete instructions prior to applying power or troubleshooting the equipment. Follow the procedures step by step.

WARNING - PERSONAL INJURY HAZARD: Read and observe all WARNING, CAUTION and NOTE labels posted on the equipment.

WARNING - BURN AND SHOCK HAZARD: Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the Power Line Monitor or associated equipment.

CAUTION: Installation wiring must be in accordance with the National Electrical Code and be consistent with all applicable local codes.

NOTE: *Read all WARNINGS, CAUTIONS and NOTES in the pertinent Instruction Book (GEK) prior to installing or removing any printed circuit card.*

RECEIVING, STORAGE AND INSTALLATION GUIDELINES

RECEIVING

The Power Line Monitor should be placed under adequate cover immediately upon receipt as packing cases are not suitable for outdoor or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may requested from GE DRIVE SYSTEMS, ERIE, PA. When seeking assistance, please use serial number, requisition number and model number to identify the equipment. (Telephone 814-875-3851)

STORAGE

If the Power Line Monitor is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures ranging from -20 degrees C (-4 F) to 55 degrees C (131 F). The surrounding air must be free of chemically corrosive or electrically conductive contaminants.

INSTALLATION

The Power Line Monitor should be installed in a well ventilated area with ambient temperatures ranging from 0 degrees C (32 F) to 40 degrees C (104 F) and relative humidities up to 90%.

IMPORTANT: *Neutral connection for 3-wire systems must be made in accordance with one of the diagrams shown in Figure 1. If neutral is not connected, the overvoltage (OVFLT) and undervoltage (UVFLT) faults will* ***not*** *function properly.*

CAUTION: Installation wiring must be in accordance with the National Electrical Code and be consistent with all local codes.

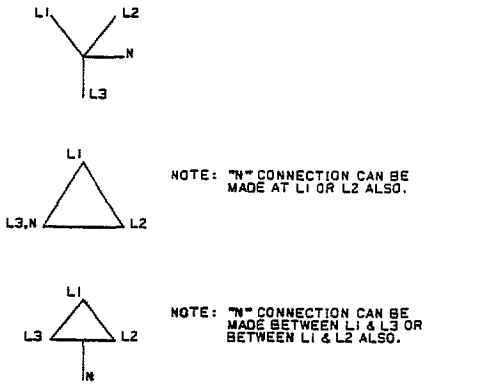


FIGURE 1. Connections For 3-Wire Systems

POWER LINE MONITOR FUNCTION AND COMPONENTS

The Power Line Monitor is a multifunctional assembly with voltage and frequency meters for monitoring a 3-phase AC power line (see Figure 2). The PLM provides transient overvoltage protection by line-to-line connected Metal Oxide Varistors (MOVs). The assembly features two printed circuit cards, the MOV card (531X128HMS_G1,G2) and the Power Line Monitor card (531X179PLM_G1). Detectors for undervoltage, overvoltage and ground faults for resistance grounded circuits are provided. Three independent ground fault detectors are available along with an MOV fuse failure detector. Several different output combinations can be selected by varying the Power Line Monitor card jumper positions (See TABLE 6). An optional enclosure is available for this assembly and is supplied with a window for viewing the meters and LEDs.

POWER REQUIREMENTS

Operation of the Power Line Monitor requires an external 115 VAC supply input. A power supply transformer, mounted on the printed circuit card, provides filtered +/-15 VDC power for the internal circuitry. The 115 VAC primary of the power supply transformer is internally fused. A single fixed voltage (8.9 VDC) is internally generated and used to reference the various trip thresholds and the frequency measurement circuit. Either 24 VDC or 48 VDC must be supplied to energize the internal relay coils and logic level fault signals. The 3-phase input to be monitored is rectified and resistance isolated from the Power Line Monitor's internal voltage levels. Input may be from either a three wire or four wire system. If a three wire system is to be monitored, the neutral input must be jumpered to one of the three incoming lines (See Figure 1). In either case, the 3-phase input is rectified, scaled and filtered to give a DC voltage proportional to the AC input voltage. This DC output is directed to the undervoltage/overvoltage circuits and a meter driver. The Power Line Monitor can accept 230 or 460 VAC systems and is set by a switch position (SW1-1 and SW1-2). (See TABLE 5 for the appropriate positions.)

UNDERVOLTAGE AND OVERVOLTAGE DETECTORS

The undervoltage and overvoltage detectors provide latching trip outputs. The thresholds are adjustable by the P2 (overvoltage) and P3 (undervoltage) potentiometers. The thresholds can be measured at the OV and UV test points (respectively) while adjusting. Each detector has an LED indicator and drops out a single set of relay contacts when tripped. A logic level output is also available from each detector in lieu of the relay contacts if selected by the appropriate jumper positions (See Table 6).

GROUND FAULT DETECTORS

The ground fault inputs are resistance isolated from the Power Line Monitor's circuits and are referenced to earth ground. The voltage signal from the externally mounted grounding resistor is rectified and filtered, then sent to a comparator that has a fixed threshold. Ground fault sensitivity is determined by selection of the grounding resistor. Each of the three detectors has an LED fault indicator and an open collector transistor to indicate ground faults. The ground fault detector outputs are not latching and can be OR'd to the input of an integrator. This integrator has an adjustable integration rate (P4) and provides a time delay for dropping out the Power Fault Relay (PFR). The Power Fault Light (PFL) LED indicates when PFR is dropped out.

MOV FUSE FAILURE DETECTOR

Three MOV fuses, a fuse detector relay and a red LED fuse failure indicator are mounted on the MOV card (see Figure 3). The LED turns ON and the relay picks up when MOV fuses fail. A circuit on the Power Line Monitor card is provided to monitor the MOV fuse detector relay. A second LED fault indicator (MOVFU) is provided on the PLM card along with a logic level output (MOVFL). The MOV failure detector can be OR'd with the ground fault outputs to drop out PFR.

CONNECTORS, MEASUREMENT POINTS, AND LED INDICATORS

CONNECTORS (Terminal Boards and Plugs)

TERMINAL BOARD GTB* is the connection for the various power points that are to be monitored. (Even-numbered points are skipped for voltage clearance purposes.)

TABLE 1: TERMINAL BOARD GTB

Point	Mnemonic	Description
1	N	Inputs to voltage monitoring and frequency readout circuits.
3	L1	(If not a four wire circuit, N may be connected to any "L".)
5	L2	
7	L3	
9	GND	Earth reference for ground fault detectors.
11	N3	Neutral to #3 ground fault detector.
13	N2	Neutral to #2 ground fault detector.
15	N1	Neutral to #1 ground fault detector.

**Terminal Board GTB was labeled 1TB on early production PLM cards.*

TERMINAL BOARD HTB* AND PLUG 1PL have like numbered points and are connected together. They contain the 115 VAC input, logic level input, MOV fuse relay contact input, and the various logic level or fault relay contact outputs.

TABLE 2: TERMINAL BOARD HTB* AND PLUG 1PL

Point	Mnemonic	Description
1	AC2	Points 1 and 2 are the 115 VAC input to on-card
2	AC1	transformer.
3	COM	Points 3 and 4 are the common for all control circuits, logic
4	COM	level inputs/outputs and reset input.
5	RST	Reset input (apply +15 VDC to reset).
6, 7, 8, 9, 10, 11, 12, 13		Points 6 thru 13 functions may vary depending on settings of jumpers JP1 thru JP8. (See TABLE 6: JUMPER DESCRIPTION.)
14	LGV	24/48 VDC input to logic and relay coils
15	NC	NOT USED

**Terminal Board HTB was labeled 2TB on early production PLM cards.*

PLUG VMPL is a two point plug to connect the voltage readout instrument. One post is at common and the other is positive with respect to common and proportional to the voltage.

PLUG FMPL is a two point plug to connect the frequency readout instrument. One post is at common and the other is a negative signal proportional to frequency.

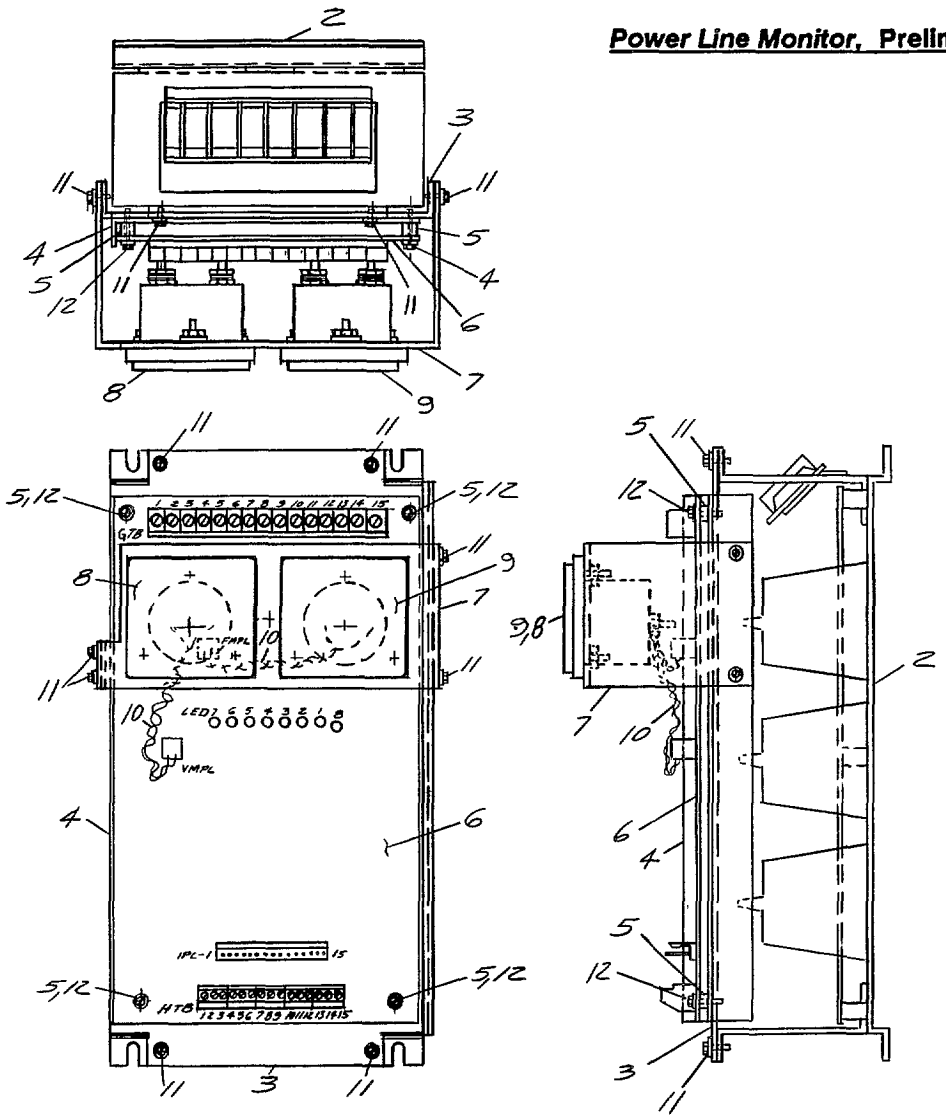


FIGURE 2. Power Line Monitor
(36C774374AAG01, 460 Volt)
(36C774374AAG02, 230 Volt)

PARTS IDENTIFICATION

Item #	Description
1	Optional Enclosure (Not Shown)
2	Base, Power Line Monitor
3	Plate
4	Insulated Support
5	Spacer
6	Power Line Monitor Card
7	Meter Panel
8	Meter, AC Voltage
9	Meter, Frequency
10	Meter Wires
11	Screw, 8-32x3/8
12	Screw, 8-32x1/2

MEASUREMENT POINTS AND LED INDICATORS

The following test posts and LED indicators are provided on the Power Line Monitor Card (531X179PLM_G1). See Figure 4 for location.

TABLE 3: TEST POSTS (531X179PLM_G1 Card)

Test Post	Description
OVT	Overvoltage threshold
UVT	Undervoltage threshold
PVT	Power Fault Relay delay circuit output

TABLE 4: LED INDICATORS (531X179PLM_G1 Card)

LED#	Color	Mnenomic	Description
1	Red	GF1	Ground Fault #1
2	Red	GF2	Ground Fault #2
3	Red	GF3	Ground Fault #3
4	Red	MOVFU	MOV fuse failure
5	Red	PFL	PFR dropped out
6	Red	OV	Overvoltage condition
7	Red	UV	Undervoltage condition
8	Green	PWR ON	115 VAC power on

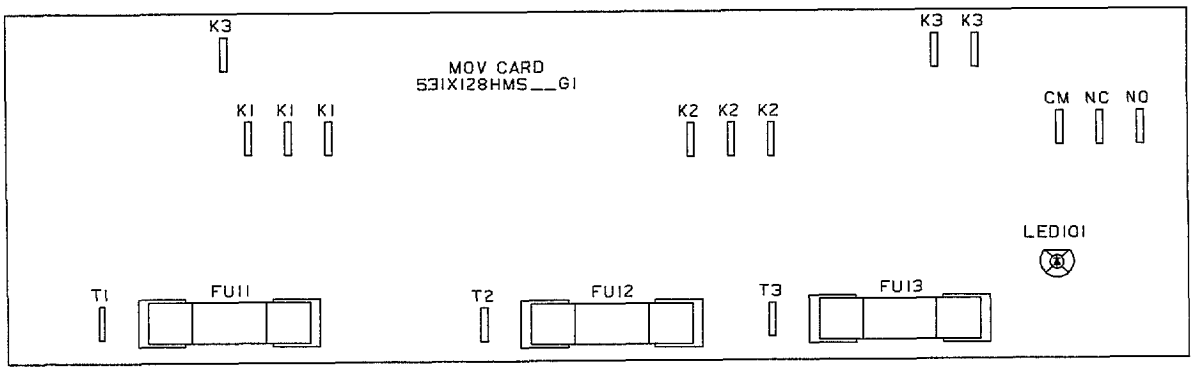


FIGURE 3. MOV Card Layout

SETTINGS AND ADJUSTMENTS

The following switch, jumper, and potentiometer settings and adjustments are provided on the Power Line Monitor card (531X179PLM_G1). See Figure 4 for location.

TABLE 5: SWITCH POSITIONS (531X179PLM_G1 Card)

Switches	Description
SW1-1 SW1-2	Switches SW1-1 and SW1-2 select the range for voltage readout, overvoltage, and undervoltage. Set switches as follows: **NOTE:** *Inadvertantly setting switches in mixed positions (one off and one on) will adversely affect meter accuracy.* **Switch** — **Position/Range** SW1-1, SW1-2 — ON - 600V full scale SW1-1, SW1-2 — OFF - 300V full scale
SW1-3 SW1-4 SW1-5	Switches SW1-3, SW1-4 and SW1-5 select the input voltage for the range select of ground faults L1, L2 and L3. Set switches as follows: **Switch** — **Position/Input Voltage** SW1-3 — ON - Ground Fault #1, 460V input SW1-4 — ON - Ground Fault #2, 460V input SW1-5 — ON - Ground Fault #3, 460V input SW1-3 — OFF - Ground Fault #1, 230V input SW1-4 — OFF - Ground Fault #2, 230V input SW1-5 — OFF - Ground Fault #3, 230V input
SW2-1 SW2-2 SW2-3 SW2-4	Switches SW2-1, SW2-2, SW2-3 and SW2-4 are set to ON to "OR" their respective faults to PFR relay. Set switches as follows: **Switch** — **Connection** SW2-1 — Connects Ground Fault #1 to PFR SW2-2 — Connects Ground Fault #2 to PFR SW2-3 — Connects Ground Fault #3 to PFR SW2-4 — Connects MOV Fuse Failure to PFR
SW2-5	NOT USED

TABLE 6: JUMPERS (531X179PLM_G1 Card)

Jumper#	Description
1	JP1 thru JP6 control the source for outputs on terminal board HTB* points 8 thru 13 (also
2	corresponding points on plug 1PL, points 8 thru 13).
3	**NOTE:** *Exercise care when setting the jumpers as many of them connect transistor*
4	*switches to various terminals of HTB*.*
5	
6	Placing JP1 thru JP6 in the 1-2 position connects relay contact outputs to HTB*(8) thru HTB*(13) as follows:

JP#	Position	Connection
1	1-2	PFR relay contacts to HTB/1PL (13)
2	1-2	PFR relay contacts to HTB/1PL (12)
3	1-2	OVR relay contacts to HTB/1PL (8)
4	1-2	OVR relay contacts to HTB/1PL (9)
5	1-2	UVR relay contacts to HTB/1PL (10)
6	1-2	UVR relay contacts to HTB/1PL (11)

Placing JP1 thru JP6 in the 2-3 position connects transistor switches to HTB*(8) thru HTB*(13) as follows:

JP#	Position	Connection
1	2-3	GFLT1 (ground fault #1) signal to HTB/1PL (13)
2	2-3	GFLT2 (ground fault #2) signal to HTB/1PL (12)
3	2-3	GVFLT3 (ground fault #3) signal to HTB/1PL (8)
4	2-3	MOVFL (MOV fuse) signal to HTB/1PL (9)
5	2-3	OVFLT (overvoltage) signal to HTB/1PL (10)
6	2-3	UVFLT (undervoltage) signal to HTB/1PL (11)

7 JP7 and JP8 select the connections to terminal board HTB* points 6 and 7 (also
8 corresonding points on plug 1PL, points 6 and 7).

With JP7 and JP8 in the 1-2 position, there is no connection to HTB* points (6) and (7).

JP#	Position	Connection
7	1-2	No connection to HTB/1PL (7)
8	1-2	No connection to HTB/1PL (6)

NOTE: *With the above setting, LED 4 (MOVFU) will be ON and logic output (MOVFL) will be high (not sinking to common). If SW2-4 is ON, PFR will also drop out.*

With JP7 and JP8 in the 2-3 position, HTB*/1PL (6) is connected to the internal +15V supply on the card and HTB*/1PL (7) is connected to the MOVFU circuit input.

JP#	Position	Connection
7	2-3	Input of MOVFU circuit to HTB/1PL (7)
8	2-3	+15V connected to HTB/1PL (6)

NOTE: *Connecting HTB/1PL (7) to HTB/1PL (8) will turn off LED4 (MOVFU) and logic output (MOVFL) will be low (sinking to common). If the relay on the MOV card is used, points CM and NC should be used to connect HTB/1PL (7) to HTB/1PL (8).*

**Terminal Board HTB was labeled 2TB on early production PLM cards.*

TABLE 6: JUMPERS (531X179PLM_G1 Card) (continued)

Jumper#	Description		
9	JP9 modifies the circuits for the voltage level being applied to LGV-HTB*/1PL (14)		
	Position	**LGV**	**Description**
	1-2	48	1.5K in series with relay coils
	2-3	24	LGV input connected to relay coils by isolating diodes.

**Terminal Board HTB was labeled 2TB on early production PLM cards.*

TABLE 7: POTENTIOMETERS (531X179PLM_G1 Card)

Pot#	Nomenclature	Description
P1	FRTRM	Calibration adjustment for frequency readout.
P2	OV	Overvoltage trip level adjustment.
P3	UV	Undervoltage trip level adjustment.
P4	DELAY	Time delay adjustment for PFR.

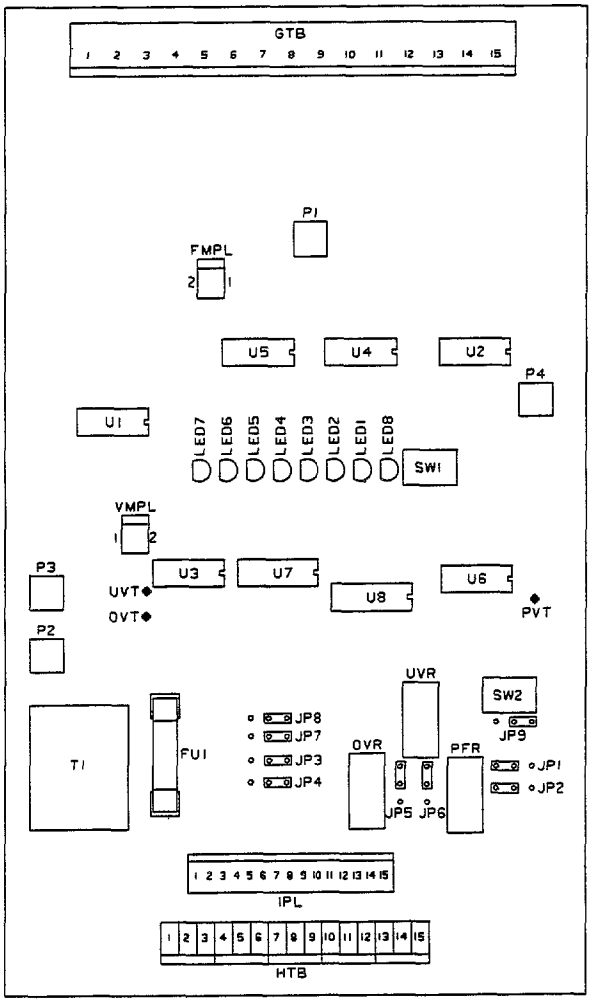


FIGURE 4. Power Line Monitor Card Layout

SET-UP AND OPERATION

WARNING - PERSONAL INJURY HAZARD: Always read the complete instructions prior to applying power or troubleshooting the equipment. Follow the procedures step by step.

WARNING - PERSONAL INJURY HAZARD: Read and observe all WARNING, CAUTION and NOTE labels posted on the equipment.

VOLTAGE READOUT CIRCUIT

When set for low voltage (300V full scale, SW1-1 and SW1-2 OFF), the nominal gain is 0.0219.
When set for high voltage (600V full scale, SW1-1 and SW1-2 ON), the nominal gain is 0.01095. (Input voltage is to be balanced and sinusoidal.)

OVERVOLTAGE TRIP CIRCUIT

The range of input voltage, based on 230/460V line, is 241/482V to 282/564V (105% to 123% respectively). This range measured at OVT will be 5.08 to 6.45 volts.

NOTE: *The overvoltage trip level must be set in the untripped state (LED6 OFF).*

Set the overvoltage trip level as follows: (See Example)

a. Determine the desired input trip level voltage.
b. Multiply the input trip level voltage by the applicable gain factor (.0219 or .01095) to obtain the desired OVT test point voltage.
c. Measure voltage at OVT with respect to common (use a high impedance input instrument such as a DVM).
d. Adjust OVT voltage with P2 as required.

EXAMPLE: +270V (Trip Level) x .0219 (Applicable Gain) = 5.91 (Volts at OVT).
Adjust P2 to obtain 5.91 Volts at OVT.

UNDERVOLTAGE TRIP CIRCUIT

The range of input voltage, based on 230/460V line, is 143/286V to 212/424V (62% to 92% respectively). This range measured at UVT will be 3.01 to 4.85 volts.

NOTE: *The undervoltage trip level must be set in the untripped state (LED7 OFF).*

Set the undervoltage trip level as follows: (See Example)

a. Determine the desired input trip level voltage.
b. Multiply the input trip level voltage by the applicable gain factor (.0219 or .01095) to obtain the desired UVT test point voltage.
c. Measure voltage at UVT with respect to common (use a high impedance input instrument such as a DVM).
d. Adjust UVT voltage with P3 as required.

EXAMPLE: 380V (Trip Level) x .01095 (Applicable Gain) = 4.16 (Volts at UVT).
Adjust P3 to obtain 4.16 Volts at UVT.

FREQUENCY READOUT CIRCUIT

The output at FMPL can be trimmed with potentiomenter P3 to be 6.0V at 60Hz. This output will be linear with frequency (+/- 3%) from 45 to 75Hz. Trimming range is approximately 20%.

GROUND FAULT DETECTORS

The trip level for the three ground fault detectors is nominally 80 volts with range switches (SW1-3, 1-4, 1-5) OFF (230V systems).
The trip level for the three ground fault detectors is nominally 160 volts with range switches (SW1-3, 1-4, 1-5) ON (460V systems).

RELAY CONTACT OUTPUTS

Normal, untripped state is with relays picked up and contacts closed. Two relays, OVR and UVR, are committed to the overvoltage and undervoltage trip circuits respectively. The third relay, PFR, is selectably driven by the three ground fault circuits and the MOV fuse failure detection circuit. The PFR driver circuit features an adjustable delay of approximately 1/4 second to 10 seconds. This delay is adjusted with P4.

Relay Contact Ratings

Card	Relay	Rating
Mov Card	K101	1.0 Amp at 125 VAC
PLM Card	UVR	1.0 Amp at 125 VAC
	OVR	1.0 Amp at 125 VAC
	PFR	1.0 Amp at 125 VAC

SPARE AND RENEWAL PARTS

Renewal parts may be obtained by contacting your local GE Sales Office or GE Drive Systems Authorized Distributor. Service may be obtained by contacting your area GE Industry Services Engineering Office.

NOTE: *Check old part to verify catalog number before ordering renewal part.*

Part Identification	Catalog Number
Meter, Frequency	104X117CC O35
Meter, AC Voltage	104X117CC 036
Power Line Monitor Card	531X179PLM_G1
MOV Card, 230 Volt	531X128HMS_G2
MOV Card, 460 Volt	531X128HMS_G1
Fuse, Special (3)	104X109AD 076
MOV (3)	104X125AD 018
Relay, Special	104X166AA 065

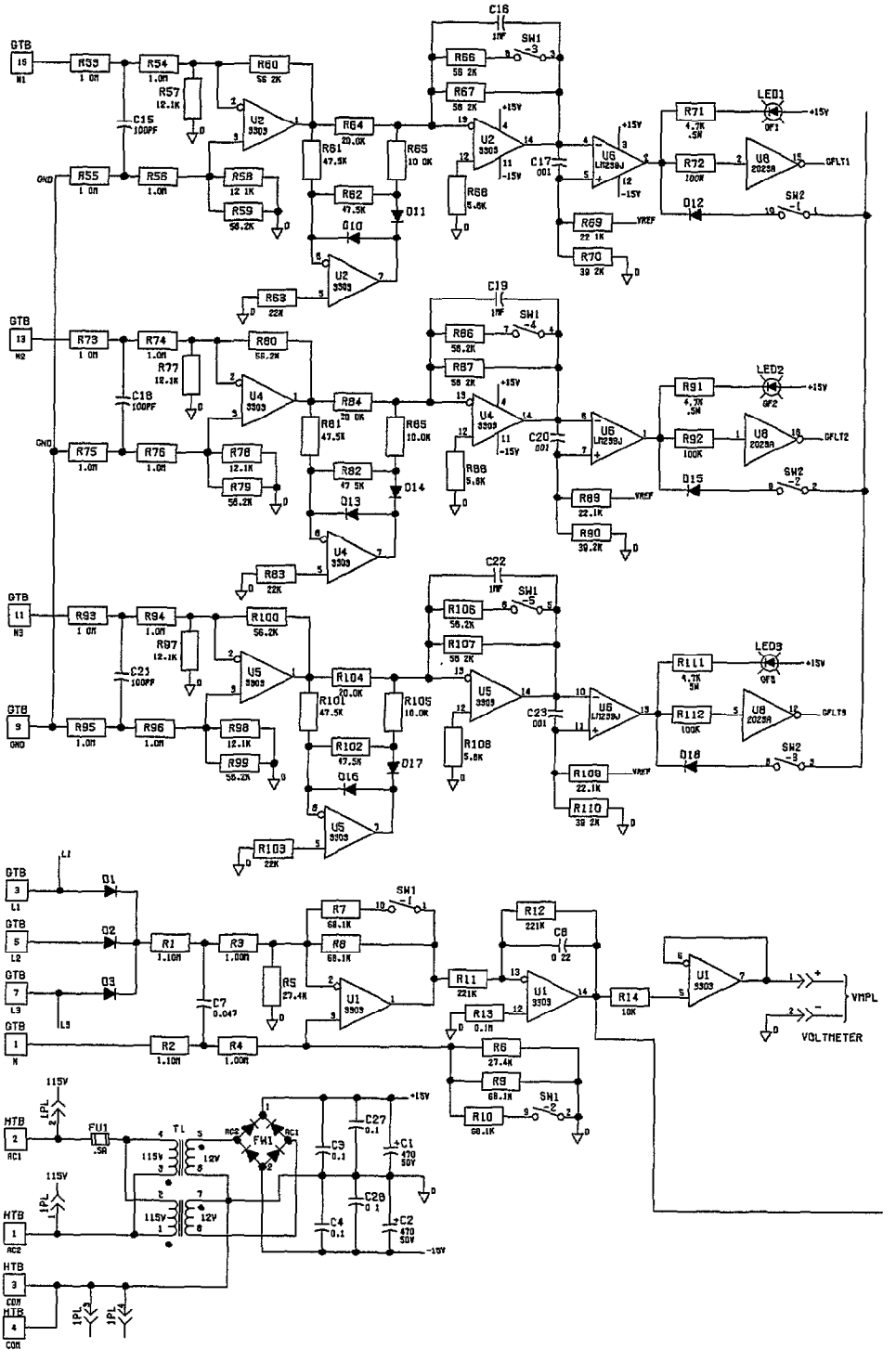


FIGURE 5. Elementary Diagram, Power Line Monitor (36D71042AH) - Part 1 of 2

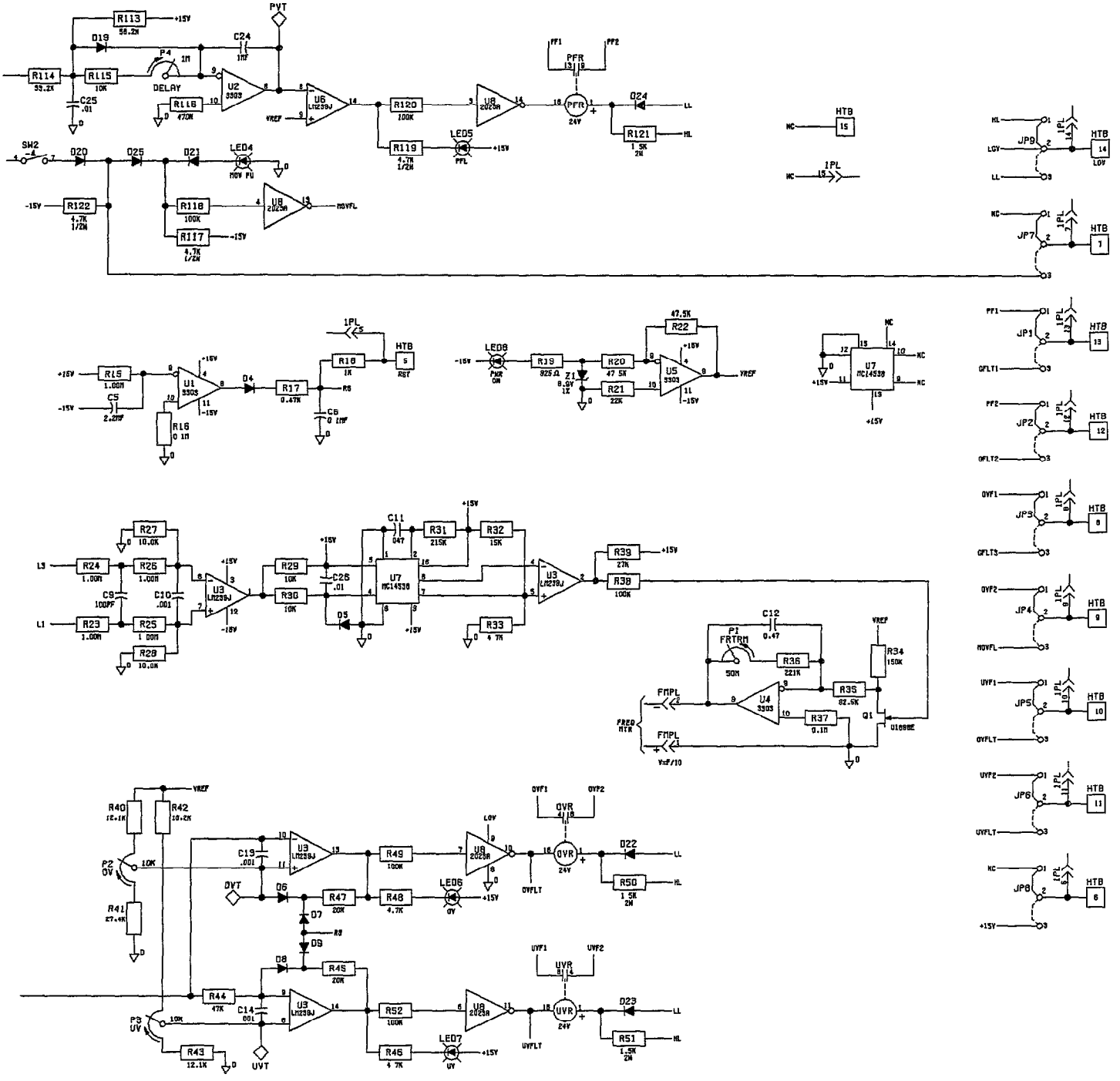


FIGURE 6. Elementary Diagram, Power Line Monitor (36D71042AH) - Part 2 of 2



GE DRIVE SYSTEMS

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Preliminary GEK-24968 (1/90)